Bytebeam Automotive BLE–CAN Module – Rev A (pre-fabrication design review)

TCAN4550-Q1 CAN FD + CC2340R5-Q1 BLE | 9–36V vehicle supply (survives reverse battery, centrally-suppressed load dump 58V) | 1S Li-ion backup provision (BQ24074)

Sheets: 1 input protection | 2 6V0 buck | 3 battery + 3V3 | 4 CAN FD | 5 MCU/BLE/RF | 6 debug/test/mechanical. Architecture diagram: docs/architecture.svg

NOTE: the assignment names 'CC2340R40'; no such part exists in TI datasheets SWRS272F/SWRS292A – the AEC-Q100 CC2340R52E0WRHBRQ1 (CC2340R5-Q1 family) is used.

1_Power_Input

File: pwr_input.kicad_sch

2_Buck_6V0

File: pwr_buck.kicad_sch

3_Battery_3V3

File: pwr_battery.kicad_sch

4_CAN_FD

File: can_fd.kicad_sch

5_MCU_BLE_RF

File: mcu_ble.kicad_sch

6_Debug_Test_Mech

File: debug_test.kicad_sch

Bytebeam hardware design assignment – pre-fabrication design review release

Sheet: /

File: bytebeam_ble_can.kicad_sch

Title: Bytebeam Automotive BLE–CAN Module (TCAN4550 + CC2340R5-Q1)

Size: A3	Date: 2026-09-29	Rev: A
KiCad E.D.A. 10.0.6		Id: 1/7

A. J1 harness connector, fuse, TVS

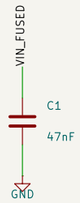
Design intent (source: LM74700-Q1 SNOSD17G, SMDJ36CA datasheet):
- Reverse battery to -36V: bidirectional TVS (VRWM 36V) does not conduct; LM74700 keeps FET off, body diode blocks.
- Load dump: TVS clamps VIN_FUSED to <= 58.1V (Vc max) - inside LM74700 ANODE 60V recommended window; FET is 80V.
- Fuse F1 protects harness/board against TVS failed-short and FET/buck faults. J1-3 IGN is fed to TCAN4550 WAKE via R-C network (CAN sheet).

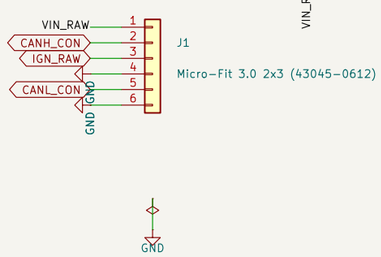




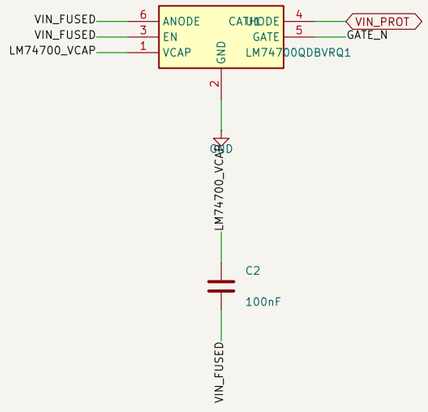


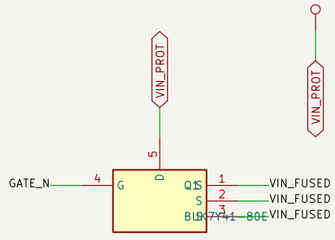


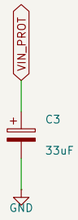


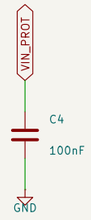


B. Ideal-diode reverse-battery protection (LM74700-Q1 + BUK7Y41-80E)





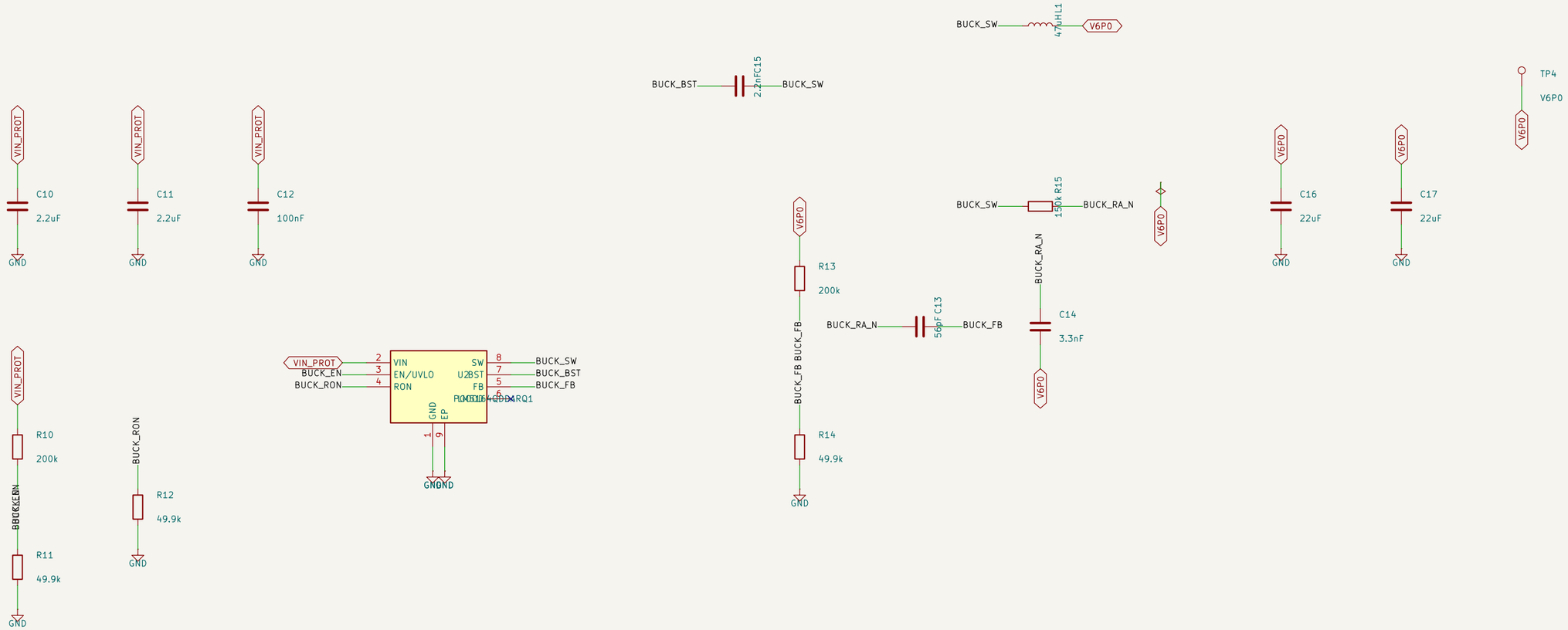






Page 2 - generated design source: tools/build_all.py
Bytebeam hardware design assignment - pre-fabrication design review release
Sheet: /1_Power_Input/
File: pwr_input.kicad_sch
Title: 1 - Vehicle supply input: connector, fuse, load-dump TVS, ideal-diode
Size: A3 | Date: 2026-09-29 | **Rev: A**
KiCad E.D.A. 10.0.6 | Id: 2/7

LM5164-Q1 COT buck: VIN_PROT (7.5–58V) -> V6P0 (6.0V, 1A)



Buck design (TI LM5164-Q1 SNVSB51C):
- VOUT=6.0V chosen so TCAN4550 VSUP (5.5–30V rec., abs 42V) stays in range and TCAN LDO dissipation is low (VSUP−5V=1V).
- fsw ~300kHz (RON 49.9k), L=47uH -> dIL ~ 0.3A pk-pk @24V; ILIM 1.5A typ. Type-3 ripple injection RA/CA/CB.
- UVLO 7.5V/7.0V: below this the buck stops; system continues from backup battery (see sheet 3).
- EMC/EMI: 300kHz harmonics fall in AM band -> CISPR25 conducted-emission test and input filter tuning are TO BE VALIDATED.

Page 3 — generated design source: tools/build_all.py

Bytebeam hardware design assignment — pre-fabrication design review release

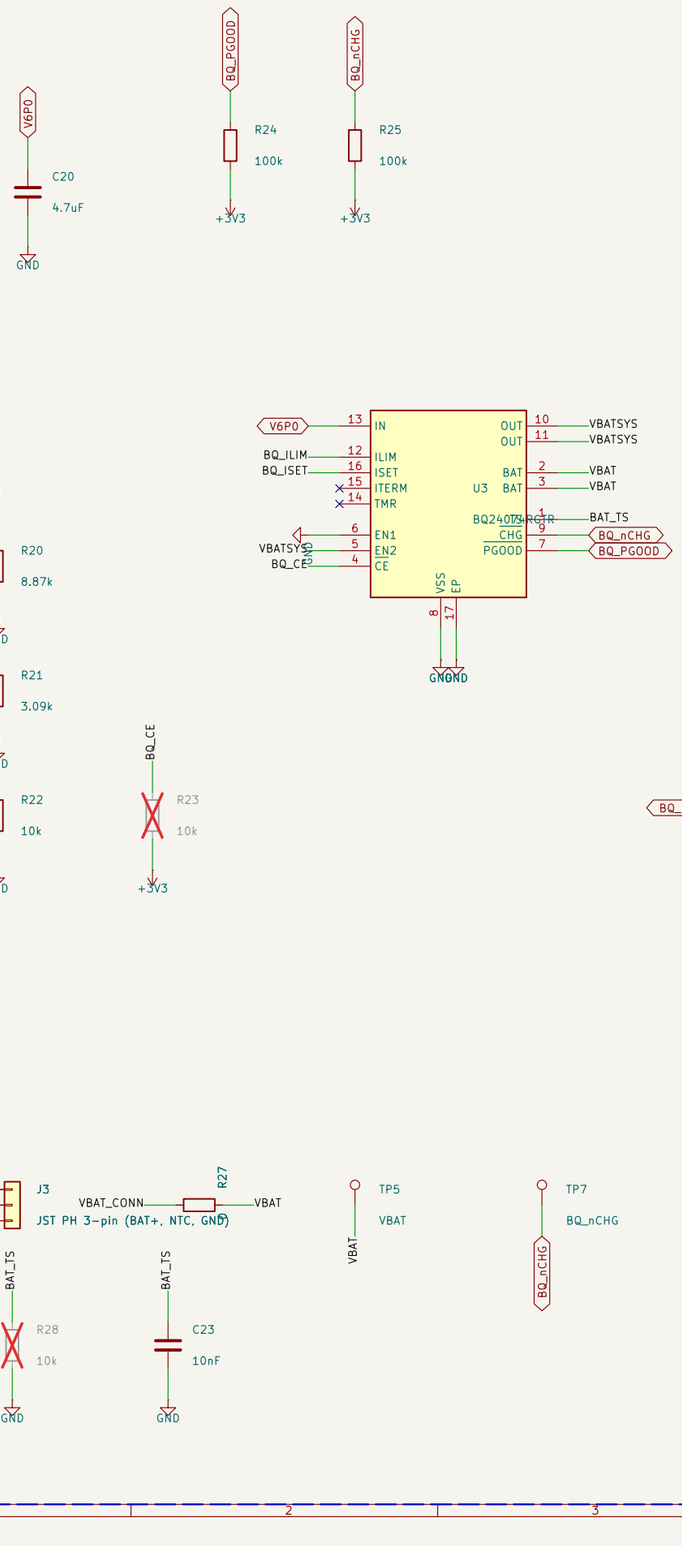
Sheet: /2_Buck_6V0/
File: pwr_buck.kicad_sch

Title: 2 — 9–36V to 6.0V wide-input buck (LM5164-Q1)

Size: A3	Date: 2026-09-29	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 3/7

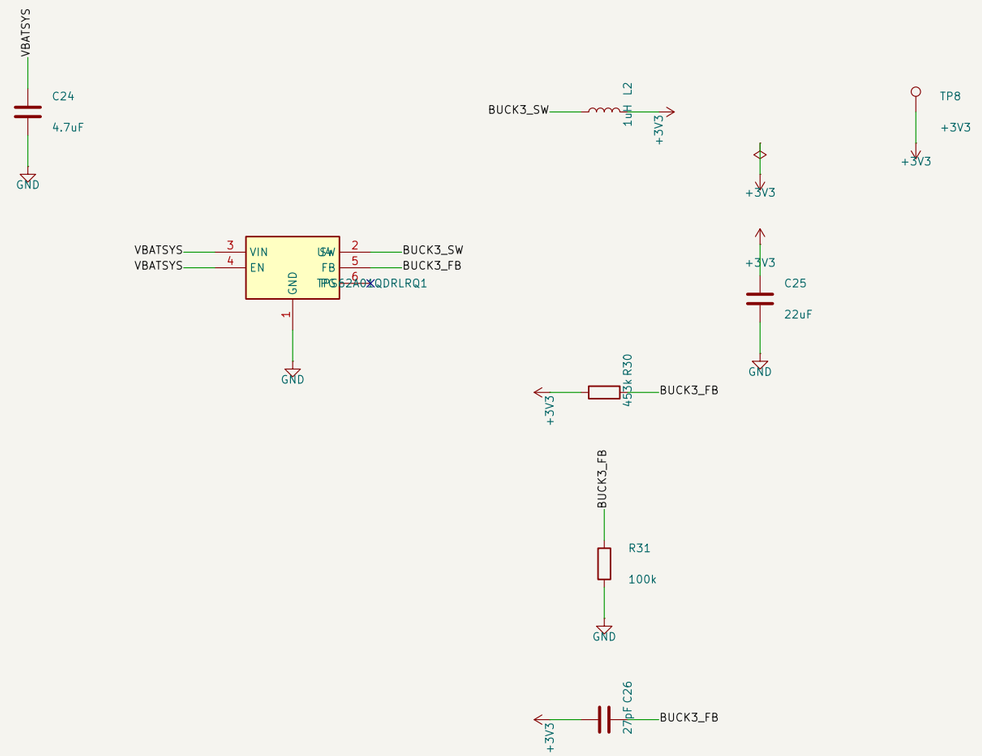
Backup battery: charger + power path

Power-path behaviour (TI BQ24074 SLUS810N):
V6P0 present: OUT regulated to 4.4V, battery charged at 100mA (ILIM 0.5A total). V6P0 lost: OUT connects to VBAT with no glitch;
TPS62A01 keeps +3V3 up -> BLE/MCU keep running; TCAN4550 loses VSUP (bus off) and is held in reset by MCU.



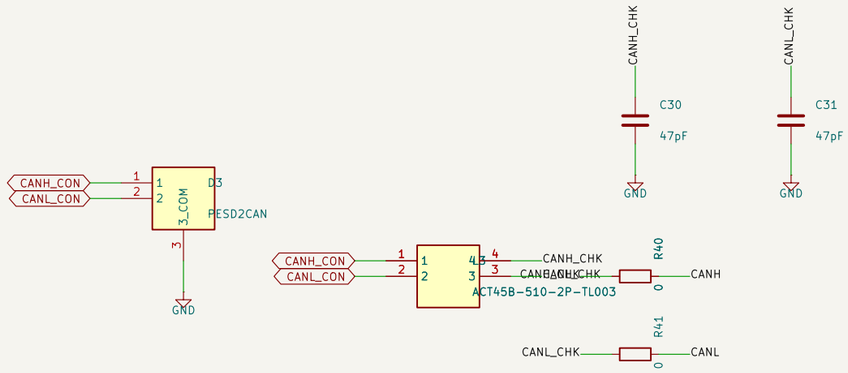
3.3 V logic rail

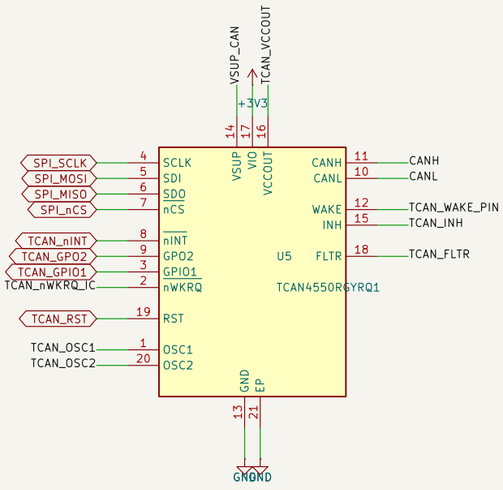
3V3 loads (typ/max): CC2340 ~7mA TX/RX / 12mA; TCAN VIO 3mA (crystal); LEDs ~2mA -> < 60mA.
TPS62A01 1A capable; large margin. Output cap 22uF (matrix ++ in datasheet Table 8-3).

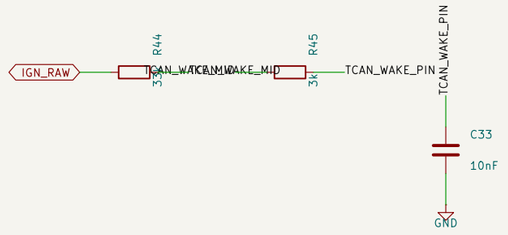


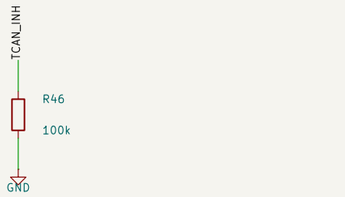
Page 4 — generated design source: tools/build_all.py
Bytebeam hardware design assignment — pre-fabrication design review release
Sheet: /3_Battery_3V3/
File: pwr_battery.kicad_sch
Title: 3 — Backup battery charger / power path (BQ24074) and 3.3V rail (TP
Size: A3 | Date: 2026-09-29 | **Rev: A**
KiCad E.D.A. 10.0.6 | Id: 4/7

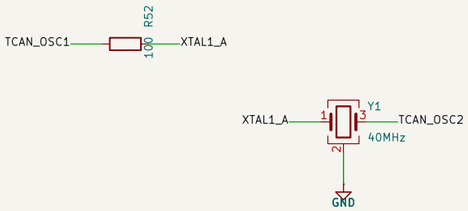
TCAN4550-Q1 CAN FD controller+transceiver (SPI to CC2340R5-Q1)

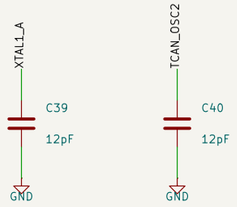




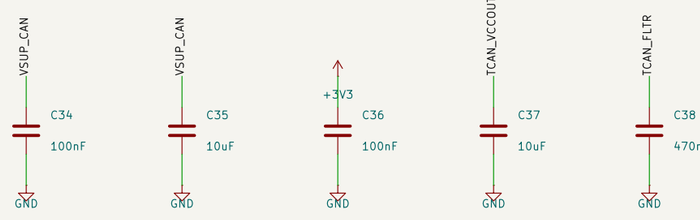












CAN physical layer robustness (TCAN4550-Q1 SLLSEZ5D, PESD2CAN, ACT45B):
ESD/surge: TCAN bus pins +/-58V DC, IEC 61000-4-2 +/-8kV contact (device level); external PESD2CAN + 51uH CMC at connector.
Bus short/open: TCAN handles CANH/L shorts to GND/supply (see DFMEA). Termination: default none (tap): JP1/JP2 close for split-120R.

TP9 CANH
TP10 CANL
TP11 TCAN_INH
TP12 SPI_MISO

Page 5 – generated design source tools/build_all.py
Bytebeam hardware design assignment – pre-fabrication design review release

Sheet: /4_CAN_FD/	
File: can_fd.kicad_sch	
Title: 4 – CAN FD interface: TCAN4550-Q1, bus protection, split termination	
Size: A3	Date: 2026-09-29
KiCad E.D.A. 10.0.6	Id: 5/7
	Rev: A

CC2340R5-Q1 core, supplies, clocks, SPI/GPIO

+3V3
+3V3_MCU
R59
0R (load placeholder)
+3V3_MCU
C50
100nF
C51
100nF
C52
100nF
C53
100nF
C54
10uF
R60
100k
nRESET
C58
1nF
VIN_PROT
R62
316k
VIN_SENSE_N
R63
27.4k
R64
10k
VIN_SENSE
C63
10nF
C55
1uF
C56
10uF
C57
100nF
VDDR
VDDD
DCDC
L4
10uH
U6
CC2340R5-Q1RGZR
RSTN
DIO16_SWDIO
DIO17_SWDCK
DIO8_SPI0SCLK
DIO11_SPI0CSN
DIO12_SPI0POCI
DIO13_SPI0PICO
DIO14
DIO18
DIO19
DIO1_A4
DIO5_A2
DIO20_A11
DIO21_A10
DIO22_A9
DIO23_A8
DIO24_A7
DIO6_A1
DIO3_X32P
DIO2_X32N
X48P
X48N
ANT
RFGND
EGP
nRESET
SWDIO
SWDCK
SPI_SCLK
SPI_nCS
SPI_MISO
SPI_MOSI
TCAN_nINT
TCAN_RST
TCAN_nWKRQ
TCAN_GPIO1
TCAN_GPO2
BQ_PGOOD
VIN_SENSE
BQ_nCHG
LED_DRV
X32P
X32N
RF_ANT
Y2
48MHz
C59
DNP
C60
DNP
Y3
32.768kHz
C61
15pF
C62
15pF
R61
1k
LED_A
D4
Green (status)

RF chain

RF design intent (SWRA834A, SWRU120D):
ANT(pin31) -> LC low-pass -> series C -> pi-match (ANT1/2/3)
-> 50R microstrip -> IFA feed. IFA at board edge,
no copper/parts in keepout, plastic enclosure.
ALL RF values = starting points, TUNE ON VNA.

RF_ANT
C70
1.2pF
L5
2.8nH
RF_F
C71
0.7pF
C72
12pF
RF_M
C73
1.5pF
C74
1.5pF
RF_IFA
R65
0
R66
0
RF_TEST
J4
U.FL (RF test)
AE1
FEED
IFA 2.4GHz
R67
0
R68
0

Page 6 – generated design source: tools/build_all.py
Bytebeam hardware design assignment – pre-fabrication design review release
Sheet: /5_MCU_BLE_RF/
File: mcu_ble.kicad_sch

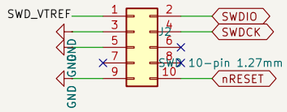
Title: 5 – BLE MCU CC2340R5-Q1, clocks, RF front-end, antenna, monitorin

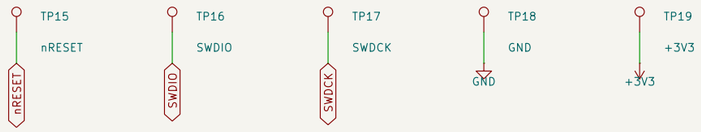
Size: A3	Date: 2026-09-29	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 6/7

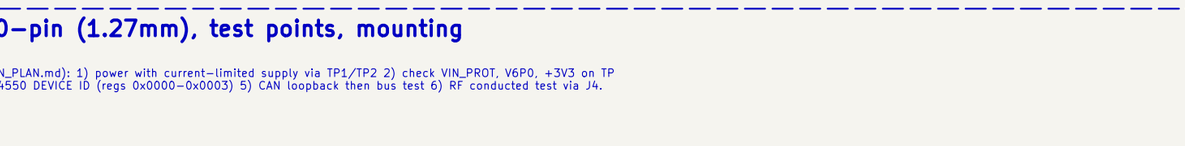
SWD/Cortex-Debug 10-pin (1.27mm), test points, mounting

Bring-up sequence (see docs/VALIDATION_PLAN.md): 1) power with current-limited supply via TP1/TP2 2) check VIN_PROT, V6P0, +3V3 on TP 3) SWD attach (J2) 4) SPI read of TCAN4550 DEVICE ID (regs 0x0000-0x0003) 5) CAN loopback then bus test 6) RF conducted test via J4.









Page 7 — generated design source: tools/build_all.py

Bytebeam hardware design assignment — pre-fabrication design review release

Sheet: /6_Debug_Test_Mech/
File: debug_test.kicad_sch

Title: 6 — Programming/debug (SWD), test points, mounting, fiducials

Size: A3	Date: 2026-09-29	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 7/7